

MINDI TAN

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EDUCATION

Texas A&M University | College Station, TX

Dec. 2027

Bachelor of Science in Computer Engineering, Minor in Mathematics

EXPERIENCE

McKim & Creed, Inc. (Signature Automation LLC)

Present

Electrical & Instrumentation Intern - Water/Wastewater E/I

Dallas, TX

- Produced AutoCAD electrical and instrumentation drawings for water/wastewater facilities, including one-line diagrams, panel layouts, and SCADA architecture to support contractor bidding and construction
- Developed PLC control logic and HMI process graphics for treatment operations, enabling reliable automatic/manual control and clearer operator visibility into system status
- Authored technical specifications and researched instruments/equipment for control system designs; supported commissioning with contractors and plant staff to validate I&C startup and minimize operational disruption

NASA Gateway to Blue Skies

March 2025

Proposal Developer and System Engineer Contributor

College Station, TX

- Co-developed SABS, a stratospheric blimp concept to harvest atmospheric water and deliver it to arid farmland; contributed to systems architecture, feasibility analysis, and multidisciplinary proposal documentation
- Developed system design schematics and proposal visuals, including two-blimp architecture, liquid-desiccant water absorption, and hydrogen ventilation analysis (8 m³/s airflow) supporting safety and materials trade studies

TAMU WIRED Robotics - Autonomous Underwater Vehicle

Aug. 2025

Electrical Team Member

College Station, TX

- Designed and integrated AUV electrical subsystems, including power distribution, wiring organization, and hardware placement within waterproofing and chassis constraints
- Partnered with mechanical and programming teams to align electrical design with sensor placement, motor control, and autonomy requirements; tested and debugged hardware during system integration to improve underwater reliability

PROJECTS

AI Smart Glasses | ESP32, INMP441, BLE, WiFi, Python/Flask, Arduino/C++

June 2026 - Present

- Designed a wearable ESP32-based system integrating an INMP441 I2S MEMS microphone, SSD1306 OLED, and LiPo power for on-glasses audio capture and subtitle display under size/power constraints
- Implemented Arduino/C++ firmware for I2S audio acquisition, I2C OLED rendering, and BLE/WiFi data transfer, enabling real-time edge I/O between the glasses and a host backend
- Built an edge-cloud speech pipeline where the ESP32 streams I2S audio over WiFi/BLE to a Python/Flask backend for speech-to-text and translation, then returns text packets to the MCU for low-latency OLED subtitle display

Interactive 3D Engineering Portfolio | Blender, Three.js, Node.js, HTML/CSS, glTF

June 2026

- Designed and modeled a custom 3D engineering workspace, constructing individual scene assets through mesh modeling, object scaling, transform application, UV unwrapping, normal correction, material setup, and hierarchical scene organization
- Built and deployed an interactive web-based 3D portfolio using Three.js, Node.js, and HTML/CSS, implementing scene loading, browser rendering, lighting, camera navigation, and interactive visualization of the modeled environment

Sharetea POS System | Java, JavaFX, PostgreSQL, Git, DAO

May 2026

- Built a full-stack POS with role-based cashier/manager clients and a PostgreSQL-backed order/inventory data model, using a DAO data layer for menu, orders, inventory, and transactions
- Implemented manager analytics pipelines for sales summaries, restock thresholds, hourly X-Reports, and end-of-day Z-Reports, aggregating live transaction data through the API and deploying the system on Render

TECHNICAL SKILLS

Languages: C++, C, Python, Java, SQL, MATLAB, Verilog, ARM Assembly, HTML/CSS

Embedded/Systems: ESP32, Arduino/C++, I2S, I2C/SPI/UART, BLE, Sensor Integration

Electrical/Hardware: Circuit Design & Analysis, Power Interfaces, PCB Assembly, Soldering, Oscilloscopes, Multimeter, Logic Analyzers, SPICE/Multism

Software/Tools: Git/Github, Linux, VS Code, Flask, AutoCAD, Xilinx Vivado

Design/CAD: Blender, Three.js, glTF, AutoCAD